

Koide Amplitude $d/c = \sqrt{2}$ and the T^4/Z_3 Geometry

Z_3 Equal-Distribution Condition, Galois Coupling Matrix (Doc. 348) and the Open Assignment to the Amplitude

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Status Markers

- [K]** *Confirmed* — derived from ξ , numerically verified.
- [B]** *Proven* — algebraically demonstrated.
- [S]** *Speculative* — open.
- [Q]** *Source* — external primary source, verified here.

Abstract

The Koide condition $Q = 2/3$ is equivalent to the amplitude condition $d/c = \sqrt{2}$ in the Z_3 circulant mass operator (Docs. A110, 352). This document examines whether $d/c = \sqrt{2}$ can be derived from the T^4/Z_3 geometry without using Koide or the lepton masses as input.

The key is the trace bilinear form of $GF(27)^*$ (Doc. 348 Theorem C): the coupling matrix between the neutrino orbits (f_1) and the lepton orbits (f_3) has normalised entries $|M_{ij}^{(13)}| = 1/\sqrt{2}$ at all non-zero positions. This is a consequence of the Z_3 equal-distribution: each element couples to exactly two of three partners, and the normalisation of the coupling strength forces the weight $1/\sqrt{2}$ per connection.

In the mass operator the circulant parameter d is the coupling weight between the isotropic part c and the Z_3 interaction term. The value $d/c = \sqrt{2}$ follows from this, however, only with the assignment $d/c = 1/|M|$; whether $d/c = |M|$ or $d/c = 1/|M|$ holds is not justified (with $d/c = |M|$ one would get $Q = 5/12$ instead of $2/3$). The target $\sqrt{2}$ (Koide amplitude) is right, but the derivation from the coupling matrix is not closed.

Equivalently, the Koide condition can be read as equipartition: at $d/c = \sqrt{2}$ the symmetric mode and the pair of non-trivial modes contribute equally to the mass sum, and the vector of the $\sqrt{m_k}$ lies at 45° to the axis $(1, 1, 1)$ (Foot 1994). This does not yet derive $\sqrt{2}$, but it makes the open question structural: what has to be justified is this half-and-half split.

Main findings: (1) The Z_3 equal-distribution yields the coupling weight $1/\sqrt{2}$ **[B]**; $d/c = \sqrt{2}$ follows from it only with the unjustified assignment $d/c = 1/|M|$ **[S]**. (2) Connection to the $GF(27)^*$ coupling matrix $|M^{(13)}|_{\text{norm}} = 1/\sqrt{2}$ **[B]**. (3) PDG satisfies $d/c = \sqrt{2}$ to $0.10 \cdot \xi$ (PDG 2022)

or $0.035 \cdot \xi$ (PDG 2024) **[K]**. (4) Bare FFGFT residual $+17 \cdot \xi$ explained by ε_i (101 %) **[K]**. (5) $Q = 2/3$ is equivalent to equal shares of the symmetric mode and the non-trivial pair, and to the 45° angle **[B]**. (6) The justification of this equipartition and the R113-bridge (origin of the falling ε_i structure) remain open **[S]**.

1 Starting Point

The Koide Amplitude Condition

The Z_3 circulant mass operator (Doc. A110) writes the square roots of the lepton masses as:

$$\sqrt{m_k} = c + d \cdot \cos\left(\theta + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2. \quad (1)$$

From $\sum_k \sqrt{m_k} = 3c$ and $\sum_k m_k = 3c^2 + \frac{3}{2}d^2$ it follows **[B]**:

$$Q \equiv \frac{\sum m_k}{(\sum \sqrt{m_k})^2} = \frac{1}{3} + \frac{d^2}{6c^2}. \quad (2)$$

The Koide condition $Q = 2/3$ is equivalent to **[B]**:

$$d = c\sqrt{2}. \quad (3)$$

This condition does not contain θ — Q depends exclusively on d/c (Doc. 352 orthogonality theorem **[B]**). The phase condition $\theta = 2/9$ is anchored in the geometry **[K]**; the amplitude condition $d/c = \sqrt{2}$ is open **[S]**.

Known: $1/\sqrt{2}$ in the Galois Coupling Matrix

Doc. 348 Theorem C derives from the trace bilinear form $\text{Tr}_{\text{GF}(27)/\text{GF}(3)}(g^{k+j})$ the coupling matrix between neutrino orbits (f_1) and lepton orbits (f_3) **[B]**:

$$M^{(13)} = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad |M_{ij}^{(13)}|_{\text{norm}} = \frac{1}{\sqrt{2}} \quad \text{at all non-zero entries.} \quad (4)$$

The weight $1/\sqrt{2}$ follows from the $\text{GF}(27)^*$ orbit structure. The possible connection to the Koide amplitude condition is the subject of this document.

2 Coupling Weight and Amplitude d/c

Step 1: The Z_3 Equal-Distribution Condition

The Z_3 symmetry of the torus T^4/Z_3 arranges the three lepton generations as equivalent elements of an orbit. A coupling between two Z_3 orbits (e.g. neutrinos f_1 and charged leptons f_3) has the property, via the trace bilinear form, that each element couples to exactly two of three elements of the other orbit (the coupling matrix (4) has exactly one zero entry per row).

The ℓ^2 normalisation of the coupling strength per row gives:

$$\sum_{j: M_{ij} \neq 0} |M_{ij}|^2 = 1 \implies 2 \cdot |M_{ij}|^2 = 1 \implies |M_{ij}| = \frac{1}{\sqrt{2}}. \quad (5)$$

This is the weight $1/\sqrt{2}$ from (4) **[B]**.

Step 2: Translation to the Circulant Mass Operator

In the circulant mass operator (1), c is the isotropic part (mean of $\sqrt{m_k}$) and d is the anisotropic part (amplitude of the Z_3 modulation).

Parameter identification:

- The isotropic part c corresponds to the self-coupling of an orbit — it sets the mass scale.
- The anisotropic part d corresponds to the coupling between orbits — it describes the interaction.

The weight of this coupling is fixed by the equal-distribution. The coupling matrix (4) normalised to the isotropic scale: every non-zero coupling has weight $1/\sqrt{2}$ [B].

For the ratio of the anisotropic to the isotropic part there are two possible assignments:

$$\frac{d}{c} = |M_{ij}|_{\text{norm}} = \frac{1}{\sqrt{2}} \quad \text{or} \quad \frac{d}{c} = \frac{1}{|M_{ij}|_{\text{norm}}} = \sqrt{2}. \quad (6)$$

Only the second yields the Koide amplitude $\sqrt{2}$; with the first one would get $Q = 5/12$ instead of $2/3$. Which assignment holds is not justified by the coupling structure. The target $\sqrt{2}$ is right, but the derivation does not carry as written; the choice $d/c = 1/|M|$ is open [S].

Step 3: Geometric Reading

The three lepton generations form a Z_3 orbit; the three neutrino generations form a second. The trace bilinear form of GF(27) describes how these orbits couple to each other. The first three steps are structurally forced, the fourth is not:

1. The Z_3 symmetry allows each element exactly 0, 1 or 2 non-zero couplings to the other orbit.
2. The trace bilinear form selects exactly 2 (Doc. 348 Theorem C [B]).
3. The ℓ^2 normalisation forces the weight $1/\sqrt{2}$ per coupling.
4. In the mass operator this weight appears as $d/c = \sqrt{2}$ only with the assignment $d/c = 1/|M|$ [S].

The condition $d/c = \sqrt{2}$ is therefore not yet derived as a consequence of the Z_3 equal-distribution in the T^4/Z_3 geometry; the step from the coupling weight to the amplitude remains open [S].

Step 4: Equipartition between Symmetric and Non-trivial Mode

With (1) the mass sum splits into two parts [B]:

$$\sum_k m_k = \underbrace{3c^2}_{\text{symmetric mode}} + \underbrace{\frac{3}{2}d^2}_{\text{non-trivial pair}}. \quad (7)$$

Exactly at $d = c\sqrt{2}$ both are equal, $3c^2$ each. The Koide condition $Q = 2/3$ therefore means: the Z_3 -symmetric mode and the pair of non-trivial modes each carry half of the mass sum [B]. Geometrically, the vector $(\sqrt{m_k})$ then lies at 45° to the symmetry axis $(1, 1, 1)$, since $\cos^2 \varphi = 1/(3Q) = 1/2$; this reading is due to R. Foot (hep-ph/9402242, 1994) [Q].

Equipartition, 45° and $Q = 2/3$ are three formulations of the same condition; the reformulation does not derive $\sqrt{2}$. It does, however, shift the open question from a number to a structure: why does the mass structure split half and half between the symmetric mode and the non-trivial pair? Once this is justified, the choice between $d/c = |M|$ and $d/c = 1/|M|$ from Step 2 becomes irrelevant.

The weight $1/\sqrt{2}$ of the coupling matrix (4) is not the same object. There it arises from normalising a row with two of three non-zero entries, here from the split symmetric versus non-trivial ($1/\sqrt{2} = \sin 45^\circ$). The numerical value agrees, the origin differs; whether the two routes belong together is open **[S]**.

3 Numerical Verification

Source	d/c	$d/c - \sqrt{2}$	Units ξ	Status
PDG 2022 [Q]	1.41420	-1.0×10^{-5}	$-0.098 \cdot \xi$	[K]
PDG 2024 [Q]	1.414209	-4.7×10^{-6}	$-0.035 \cdot \xi$	[K]
FFGFT bare	1.41649	$+2.28 \times 10^{-3}$	$+17.1 \cdot \xi$	[K]
FFGFT + ε_i	1.41422	$+8 \times 10^{-6}$	$+0.06 \cdot \xi$	[K]
Target (Koide condition)	$\sqrt{2} = 1.41421$	0	0	[B]

PDG satisfies the condition to $0.10 \cdot \xi$ (PDG 2022); with m_τ from PDG 2024 to $0.035 \cdot \xi$, corresponding to $Q - 2/3 = -0.017 \cdot \xi$. The bare FFGFT residual of $+17 \cdot \xi$ is fully explained by the generation-dependent corrections ε_i (Doc. 352 §9, 101 %) **[K]**. PDG masses: $m_e = 0.51100$ MeV, $m_\mu = 105.658$ MeV, $m_\tau = 1776.86$ MeV **[Q]** (PDG 2022) or $m_\tau = 1776.93 \pm 0.09$ MeV **[Q]** (PDG 2024); the ε_i rows are based on PDG 2022 (Doc. 352).

4 Status Overview

	Statement	Status
1	$Q = 2/3 \iff d/c = \sqrt{2}$ algebraically from Z_3 circulant	[B]
2	Q does not contain θ ; d/c and θ are orthogonal parameters	[B]
3	Z_3 equal-distribution of coupling to 2 of 3 gives the weight $1/\sqrt{2}$	[B]
3a	$d/c = \sqrt{2}$ from this weight (assignment $d/c = 1/ M $ unjustified)	[S]
3b	$Q = 2/3 \iff$ equipartition symmetric/non-trivial $\iff 45^\circ$ to $(1, 1, 1)$	[B]
3c	Justification of this equipartition from the geometry	[S]
4	Connection to $GF(27)^*$ coupling matrix $ M^{(13)} _{\text{norm}} = 1/\sqrt{2}$	[B]
5	PDG satisfies $d/c = \sqrt{2}$ to $0.10 \cdot \xi$ (2022) or $0.035 \cdot \xi$ (2024)	[K]
6	Bare FFGFT residual $+17 \cdot \xi$ explained by ε_i (101 %)	[K]
7	Dynamical derivation of the falling ε_i structure	[S]

The open bridge from Doc. 352 (derivation of $d/c = \sqrt{2}$ from the geometry) is therefore not closed; it remains **[S]**.

5 Note: Derivation Chain for Precise Lepton Masses

The lepton ladder is not in the chain for the precise mass derivation.

The lepton ladder ($m_i = r_i \cdot \xi^{P_i} \cdot \nu_i$, Doc. A100) yields masses at the percent level (1 %). It is a coarse framework, but not a necessary link in the precise derivation. The **[S]** entry

in Doc. 352 §7 ("Koide is not contained in the FFGFT lepton ladder") concerns the ladder alone — it is correct and remains [S]. It is, however, *not* in the chain for the precise masses.

The chain for the precise lepton masses runs exclusively through the Z_3 circulant:

1. $\theta = 2/9$ from the Z_3 circulant geometry [K] (Docs. A110, 291)
2. $d/c = \sqrt{2}$ — Koide amplitude; the derivation from the Z_3 equal-distribution condition is open [S] (this doc.)
3. Reference point m_μ/m_e [Q] — one declared empirical anchor (Doc. A110, P42)
4. $m_\tau = 1776.9690 \pm 0.0001$ MeV [K] (Docs. A110, A210)

The [S] entries in Doc. 352 concern residual analyses (ε_i structure, R113-bridge) or the ladder–Koide relation — none of these is in this chain. Within the chain itself, however, step 2 is open: $d/c = \sqrt{2}$ is not derived from the geometry [S]. Moreover, the exact pair $(\sqrt{2}, 2/9)$ is excluded for m_μ/m_e ($\approx 440\sigma$ against CODATA, Doc. 369).

Conclusion: The derivation chain for the precise lepton masses is not fully closed; the derivation of the amplitude $d/c = \sqrt{2}$ remains open [S].

6 Verification Script

- Numerical verification of the Z_3 normalisation condition, coupling matrix entries (4), d/c calculation from PDG and bare FFGFT masses, ε_i correction, equipartition, 45° angle and PDG 2024 (11 checks): `python/Dok353_Skripte/pruef_353_koide_amplitude.py`

7 Open Points

- Why do ε_i exhibit a falling generation-dependent structure ($\varepsilon_e > \varepsilon_\mu > \varepsilon_\tau$)? Attribution to QED extraction and SI projection is [S] (R113-bridge, Doc. 352 §10).
- Full derivation of CKM/PMNS mixing angles from the Galois bilinear form (Doc. 348 Theorem D) [S].
- Why does the mass structure split half and half between the Z_3 -symmetric mode and the pair of non-trivial modes (Step 4)? Only this justification would derive $d/c = \sqrt{2}$ from the geometry; the question $d/c = 1/|M|$ or $d/c = |M|$ would then be settled [S].
- Are c and d individually fixed by the geometry, or only their ratio? [S]

8 Use of AI Tools

This document was produced with the assistance of AI-based tools. The questions, postulates and all content decisions originate with the author; the tools formulate, compute and generate verification scripts. Responsibility for content and errors lies entirely with the author. The detailed wording is in A010.